

PLAN OVERVIEW

A Data Management Plan created using DMPonline

Title: Association between residential exposure to pesticides and cognitive development in children, a pilot study

Creator: Inka Pieterse

Principal Investigator: Anke Huss

Data Manager: Inka Pieterse

Project Administrator: Maartje Pronk

Affiliation: Utrecht University

Funder: National Institutes of Health (NIH)

Template: UU Data Management Plan (DMP)

Project abstract:

People who live in the vicinity of agricultural land have a higher exposure to pesticides. It is yet unclear whether such exposure leads to adverse health outcomes. However, the information available is insufficient to explain these results.

As a result, the Health Council of the Netherlands issued a recommendation in 2020 to conduct further health research into the association between pesticides and the health of residents. In particular, a need was identified for greater insight into long-term effects. The Health Council of the Netherlands made its recommendation conditional on a more accurate definition of exposure to pesticides. In addition, RIVM made a recommendation to include more information on the lifestyle of local residents in the research. Subsequently, IRAS, NIVEL and RIVM compiled an overview of the feasibility and usefulness of several research options.

The presented research options include cognition of children, such as learning disadvantages as one of the relevant research questions. The suggested study has three distinct phases: A pilot project in phase 1 (2024-2026), a cross-sectional study with measurements including about 2000 children in phase 2 (2026-2029), and an analysis of the national registry (CITO) of school grades in phase 3 (2029-2030).

The current DMP addresses phase 1, the pilot study.

The overall aim of this study is to explore the methods for measuring exposure to pesticides in children and study the best way to measure cognition in children as a pilot for a larger study in the future to examine the association between exposure to pesticides and cognition in children.

In particular, in the pilot phase we want to develop all communication material, assess all exposure measurement procedures and cognitive testing procedures regarding communication challenges and online tools, and to develop recruitment options.

The aim is to do all sample and data collection remotely, without in-person contact between researchers and study participants. The associated challenges are the focus of the pilot study.

ID: 187484

Start date: 01-07-2024

End date: 01-07-2026

Last modified: 06-10-2025

Grant number / URL: N.A.

ASSOCIATION BETWEEN RESIDENTIAL EXPOSURE TO PESTICIDES AND COGNITIVE DEVELOPMENT IN CHILDREN, A PILOT STUDY

DATA COLLECTION

1.1 Will you re-use existing data ?

If yes: explain which existing data you will re-use and under which terms of use.

- Yes, I will re-use existing data

We will use as support environmental exposure assessment data, which is performed in parallel in another sub-study (protocol NL54727.041.15, 'residents and pesticides', OBO2 WP2)

1.2 Describe your data.

Fill the table below with a brief description of the data, including the type, format and volume.

#	Data Description	Data Type	Format	Total Volume
1	Formdesk data from registration form	Tabular	.csv	
2	Signed informed consent data from Validsign	Tabular	.csv	
3	Exposure data (hair, urine, house dust, wristband)	Tabular		
4	Qualtrics questionnaires parents and child (baseline, exposure measurements, feedback) and teacher (child performance)	Tabular	.csv	
5	Cognition test data from Millisecond	Tabular	.xlsx	
6	CITO score from CBS	Tabular		

DATA DOCUMENTATION

2.1 Describe the documentation and metadata that you will use to to make your data reproducible and interoperable.

Describe which files you will provide, along with a brief description of the information they will contain, to make your data reproducible and interoperable. Describe the information that you will provide to make the

data items in questions 2.1 reusable and interoperable. If using a specific metadata standard, please mention this below.

Paper versions of the forms and questionnaires together with codebooks will describe the contents of the created datasets.

Data stored in YODA: "The metadata in Yoda complies with the Datacite V4 standard. The metadata is stored persistently along with the data and in the Yoda repository. It is findable in Datacite upon publication of data. The metadata in Yoda can also be retrieved via a service that complies with the Open Archives Initiatives Protocol for Metadata Harvesting (OAI-PMH)."

2.2 Describe the folder structure you will provide to make your data reproducible and interoperable. Describe the folder structure, naming conventions and/or version control you will use for this project.

At the UU network drives a privacy folder (for the identifying data) and a project folder (for the pseudonymised research data) will be used:

O:\DGK\IRAS\EEPI\Privacy\OBO2WP6

O:\DGK\IRAS\EEPI\Projects\OBO\OBO-II Cognition

At YODA:

"We will archive and publish our data directly in Yoda. After publication, the data is:

- Findable via its DOI and the associated metadata.
- Accessible under Restricted access with specified data access conditions.
- Interoperable as made certain by review of the data manager prior to publication.
- Reusable via an explicitly defined license that is part of the metadata."

DATA STORAGE

3.1. Select the storage solution(s) where you will store and back-up your data.

Select the locations where your data will be stored. You may select more than one. Please describe the storage solution and the backup strategy of your storage solution if it does not appear in the list below.

- YoDa
- O:Drive

DATA PRIVACY AND SECURITY

4.1 Will you be collecting or using personal data ?

Personal data is any data which, alone or in combination with other information, can identify a living person. Such data must abide by the GDPR and requires additional safeguards and documentation to be processed lawfully.

- Yes, I will collect and/or use personal data

A DPIA has been created for this project by the project team in cooperation with the DGK privacy officers

4.2 What is the legal basis by which you are collecting and/or processing this data ?

If you are uncertain as to which legal basis applies to your type of research; please do not hesitate to contact us at info.rdm@uu.nl or by using the "Request feedback" button and leaving a comment alongside this question.

- Informed consent

4.3 Select the privacy and security measures you will employ to protect the privacy of your data subjects. Check all that apply.

- Encryption
- Aggregation/Abstraction
- Access control
- Minimization
- Pseudonymization
- Secure storage

4.4 Who is the controller of the personal data ?

The controller of the personal data is the entity which determines what is done with the data. In most cases the controller is Utrecht University.

Utrecht university is the controller of the collected personal data. Nevertheless, the principal investigator of the research project will ensure that the data is handled and processed in accordance with the GDPR.

4.5 How will ownership and intellectual property rights of the data be managed?

Describe who controls access to the data and who determines what is done to the data.

Utrecht University (UU) is the data controller.

UU determines the purpose and means and is therefore the data controller.

Wageningen Food & Safety Research is the processor.

The processor is responsible for the exposure analysis of the urine sample, house dust, wristbands, and hair sample. The person responsible within this group is Dr. JGJ (Hans) Mol, Senior Researcher in the field of pesticides, natural toxins, food safety, and exposure.

Formdesk, Validsign, Qualtrics, Millisecond

Parts of the services of these processors are used to conduct the study.

Agreements have been concluded with these parties.

UU has signed a data processing addendum with Qualtrics.

UU has an agreement with Formdesk regarding the secure use of personal data.

UU researchers can receive pseudonymised research data upon request after agreements have been made about this and only if the research they intend to conduct fits within the purposes for which personal data was collected.

The pseudonymised research data may also be shared with the RIVM and/or other external researchers via Surf File Sender. This is done under a data transfer agreement, which will only be concluded if the intended use aligns with the research objectives.

DATA SELECTION, PRESERVATION & SHARING

5.1 Describe the data you will be preserving and the storage solution where it will be preserved?

Describe which data will be preserved under long-term storage. You may refer back to the data described in question 1.2 to specify which data will be preserved. Explain where you will preserve your data, and how procedures are applied to ensure the survival of the data for the long term.

Data collection is scheduled to start in July 2024 and will continue until July 2026 at the latest, at which time the study will be terminated.

All data will be downloaded by the OBO Data Manager and stored on the internal UU network drives. These data will then be deleted from Formdesk, Validsign, Qualtrics, and Millisecond.

Identifying data will be stored in the Privacy folder on the UU's O-drive, to which only the project's Data Manager has access.

Other research data will be stored in the Project folder on the UU's O-drive or in YODA, where researchers will have access to the pseudonymised data for analysis.

YODA facilitates collaboration on data during research, archiving of data during and after research, and publication of research data. Data along with its metadata is shared within a closed user group, accessible to authorized users. In Yoda, data can also be deposited in the Vault where it becomes read-only. Vaulted data can be published (findable and citable via DOI Datacite identifier) and as such can be made available to the research community at large.

Directly identifying personal data will be retained on the internal UU network drives for 10 years after the researchers have published the research results. After deletion of these personal data, the remaining research data can no longer be linked to the participants and are therefore anonymized. It is necessary to retain both the research data and the directly identifying personal data for this period because research results must still be verifiable and reproducible after publication. To prevent directly identifying data from becoming inaccessible upon the departure of the project Data Manager, access will also be granted to an IRAS Data Manager not involved in the project, ensuring the data remains accessible throughout the retention period.

A different retention period applies to the following data:

Participants and their parents were asked to provide contact information in order to participate. This personal data will be removed from the data collected through the study once all components have been completed for the participant.

5.2 Describe the data you will be sharing and the repository where it will be shared?

Describe which data you will be sharing. Select where you will make your data findable and available to others. If selecting "Other" please specify below which repository and provide a URL.

Please also write below if you will apply any conditions to the re-use of your data. (i.e. Creative commons license or Data Transfer Agreement).

- YoDA

Data will be shared in the YoDA repository.

The data will be re-usable under a CC-BY creative commons license.

Sharing with external researchers is done under strict conditions regarding security and permitted use. The research must fit within the purpose of the study for which the participant has given informed consent. The sharing of the pseudonymised research data takes place with the application of a data transfer agreement. Directly identifying personal data is never shared.

5.3 Are specialized, uncommon or expensive software, tools or facilities required to use the data?

Please list any specialized, uncommon or expensive software, tools or facilities that are absolutely required to obtain, use or handle your data, if any.

No, all the data can be accessed by free, open-source or non-proprietary software

DATA MANAGEMENT COSTS AND RESOURCES

6.1 What are the foreseeable research data management costs and how do you expect to cover them ? Please specify the known and expected costs involved in managing, storing and sharing your data. Also explain how you plan to cover these costs.

Data management costs are budgeted as personnel costs and therefore part of the project budget. This is 0.2 FTE continuously dedicated to data management.

6.2 Who will be responsible for data management?

Please specify who is responsible for updating the DMP and ensuring it is being followed accordingly.

Inka Pieterse, (project) data steward of UU

6.3 State if you contacted an RDM consultant from Utrecht University to help you fill out your DMP.

Please list their name and date of contact.

This is mandatory for NWO grants.

No.